

PURDUE UNIVERSITY, IN, USA

WMO: 724386

Lat: 40.412N

Lon: 86.937W

Elev: 183

StdP: 99.15

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 14835

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-18.0	-15.1	-22.4	0.5	-17.5	-19.6	0.7	-14.5	11.7	-3.9	10.6	-3.5	3.1	270	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.8	33.0	24.1	31.8	23.5	30.3	22.6	25.8	30.9	25.0	29.7	24.1	28.5	4.0	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.3	19.7	29.1	23.5	18.7	27.9	22.6	17.8	26.7	81.0	31.0	76.6	29.7	73.1	28.7	31.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.0	8.7	7.9	-22.1	35.2	4.3	1.8	-25.2	36.5	-27.8	37.6	-30.2	38.6	-33.3	40.0		
			-22.5	27.5	4.2	1.2	-25.5	28.4	-27.9	29.1	-30.3	29.8	-33.3	30.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	11.4	-3.0	-0.7	4.9	11.4	17.0	22.0	23.5	22.9	19.4	12.8	5.8	0.0
	DBStd	10.67	7.06	6.51	6.26	5.22	4.72	3.50	3.07	2.86	4.05	5.07	5.58	6.03
	HDD10.0	1432	405	304	184	43	4	0	0	0	1	28	149	313
	HDD18.3	3089	661	534	418	215	84	9	1	2	36	184	377	568
	CDD10.0	1942	2	3	26	86	220	361	420	400	283	114	23	4
	CDD18.3	558	0	0	2	7	42	120	163	144	68	13	0	0
	CDH23.3	5021	0	0	13	65	394	1075	1498	1232	629	113	1	0
	CDH26.7	1708	0	0	2	8	109	378	554	417	216	25	0	0
Wind	WSAvg	3.4	4.1	4.1	4.2	4.4	3.4	2.8	2.5	2.4	2.6	3.3	3.7	3.8
Precipitation	PrecAvg	905	44	38	67	94	93	103	86	103	75	62	72	73
	PrecMax	1091	112	100	145	222	164	208	180	310	224	128	228	197
	PrecMin	751	2	8	10	15	25	7	16	22	0	5	8	8
	PrecStd	107	34	23	33	49	38	54	45	71	56	31	46	44
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.4	18.3	25.3	27.8	31.8	34.0	35.8	34.0	33.3	29.5	22.4	17.4
		MCWB	13.9	13.0	16.8	18.4	21.4	24.2	26.3	24.6	22.3	20.2	15.0	14.8
	2%	DB	12.9	14.9	21.4	25.3	29.5	32.3	32.9	32.4	31.2	26.6	19.1	14.1
		MCWB	11.2	11.7	14.9	17.1	20.2	23.3	25.0	24.4	21.9	18.6	14.5	11.8
	5%	DB	10.0	11.8	18.2	22.9	27.6	30.7	31.4	30.8	29.1	23.8	17.1	11.4
		MCWB	8.1	8.9	13.0	15.6	19.6	22.6	24.2	23.4	20.7	17.4	13.5	9.4
	10%	DB	6.3	8.3	15.2	20.7	25.5	28.9	29.8	28.9	27.2	21.4	14.7	8.1
		MCWB	4.4	5.7	10.7	14.5	18.7	21.6	23.1	22.4	19.7	16.3	11.2	6.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.4	15.2	17.7	20.4	23.6	25.8	27.7	26.7	24.5	22.0	17.5	15.3
		MCDB	15.8	17.3	24.3	25.8	29.0	31.7	33.0	32.0	30.8	27.1	20.3	17.3
	2%	WB	11.8	12.1	15.7	18.4	22.0	24.8	26.4	25.5	23.4	20.4	15.6	12.3
		MCDB	12.9	14.2	20.2	23.1	27.3	30.2	31.4	30.6	28.6	24.1	18.1	13.8
	5%	WB	8.2	8.9	13.9	16.8	20.8	23.9	25.3	24.6	22.2	18.7	14.0	9.3
		MCDB	9.5	11.6	17.7	21.5	25.4	28.8	29.6	29.0	26.5	22.3	16.5	10.9
	10%	WB	4.5	5.9	11.3	15.3	19.8	22.9	24.3	23.6	21.1	16.9	11.5	6.5
		MCDB	6.2	8.4	14.7	19.8	23.9	27.3	28.5	27.4	25.2	20.7	14.3	8.2
Mean Daily Temperature Range	5% DB	MDBR	8.1	8.9	10.1	11.6	11.4	10.9	10.8	10.9	12.3	11.3	9.3	7.7
		MCDBR	10.4	12.8	13.6	14.0	13.5	12.7	11.8	12.1	14.2	14.1	11.9	10.4
		MCWBR	8.8	10.1	8.7	8.1	6.7	5.7	5.3	5.4	6.0	7.4	8.4	8.8
	5% WB	MCDBR	9.9	12.0	11.8	12.0	11.3	10.8	10.5	10.4	11.6	11.5	10.6	9.8
		MCWBR	9.2	10.4	8.1	8.0	6.4	5.5	5.5	5.4	5.7	6.9	8.5	8.9
Clear-Sky Solar Irradiance	taub	0.299	0.326	0.350	0.404	0.434	0.441	0.459	0.458	0.422	0.364	0.321	0.299	
	taud	2.427	2.339	2.359	2.218	2.188	2.208	2.157	2.147	2.227	2.384	2.454	2.451	
	Ebn at Noon	865	885	901	868	847	840	820	809	813	830	829	834	
	Edh at Noon	80	101	111	137	145	142	148	146	126	96	78	73	
All-Sky Solar Radiation	RadAvg	1.73	2.60	3.64	4.71	5.38	6.16	6.11	5.46	4.53	3.09	2.04	1.44	
	RadStd	0.18	0.26	0.28	0.40	0.48	0.37	0.34	0.31	0.35	0.29	0.21	0.16	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (45 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page